

MUKUL

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PROFESSIONAL SUMMARY

AI/ML Engineer at TCS with experience developing NLP-based defect search applications and Python automation for XML and Excel workflows. Builds full-stack AI projects using hybrid retrieval, source-grounded generation, and retrieval evaluation, with a focus on reliable backend logic and practical user workflows.

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS)

June 2025 – Present

Assistant System Engineer | AI/ML Team

- Built a Flask defect-search application from scratch, with a frontend to enter defects and view similar historical records from CSV data.
- Implemented three-stage search: shortlisted 20 defects with BM25 and word/character TF-IDF, reranked using Word2Vec and WordNet, and scored the top five matches.
- Applied text normalization, lemmatization and summary weighting; filtered results using a minimum similarity threshold.
- Built Python automation to extract interface IDs, incoming/outgoing direction and other XML fields into structured Excel reports.
- Generated XML and Excel outputs from source XML using conditional transformations and specified field mappings.

PERSONAL PROJECT

ResolveIQ | Incident Resolution with Hybrid RAG

[View source code on GitHub](#)

Python, FastAPI, React, Gemini, BM25, FAISS, PostgreSQL/pgvector

- Built document ingestion for PDF, CSV and TXT, preserving page/location metadata through chunking and embedding generation.
- Combined lexical and semantic retrieval using Reciprocal Rank Fusion, with optional cross-encoder reranking, confidence gating, abstention and source citations.
- Added role-based access, an allow-listed demo and retrieval evaluation using Hit@1, Hit@5 and MRR@5; validated 50 backend unit tests.

TECHNICAL SKILLS

Languages: Python, Java, SQL

NLP and retrieval: BM25, word and character TF-IDF, cosine similarity, Word2Vec, WordNet, text preprocessing

AI application development: RAG, Gemini, embeddings, FAISS, pgvector, prompt engineering

Engineering and tools: Flask, FastAPI, React, pandas, NumPy, scikit-learn, XML, Git, Docker

EDUCATION

Bachelor of Computer Applications (BCA)

2021 – 2024

Uttaranchal University, Dehradun | CGPA: 8.3/10

Master of Computer Applications (MCA)

2026 – 2028